

```

ClassDensityPtb():
  valEdges  $\leftarrow \emptyset$ 
  d  $\leftarrow$  value  $\in [0 \dots 1]$ 
  ptbRate  $\leftarrow$  value  $\geq 0$ 
  budget  $\leftarrow$  ptbRate  $\times |G.edges|$ 

  C1, ..., Cn  $\leftarrow$  nodes with class 1, ..., n
  for Ci  $\in$  C1, ..., Cn do:
    Gi  $\leftarrow$  IndSubg(Ci)
    for cc  $\in$  ConComp(Gi) do:
      if density(cc)  $\geq d$  then
        Clique  $\leftarrow$  MakeClique(cc)
        append(Clique.edges, valEdges)
      end if
    end for
  end for

  while under budget do
    edge  $\leftarrow$  randSel(valEdges)
    if edge  $\notin G$  then
      addEdge(G, edge)
    end if
  end while

```